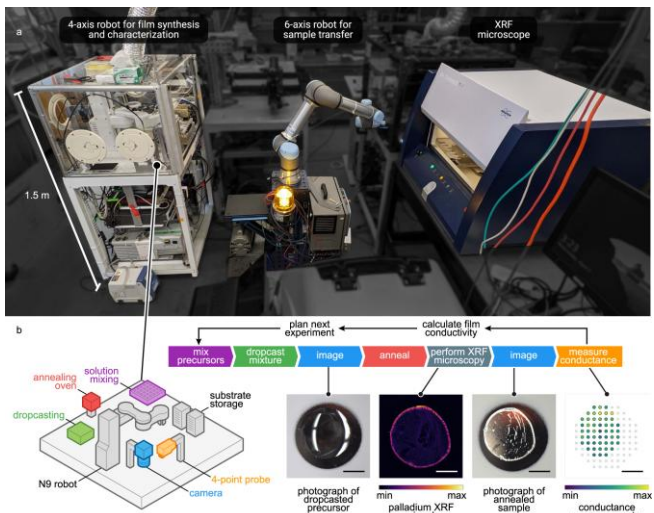


Problem 1:

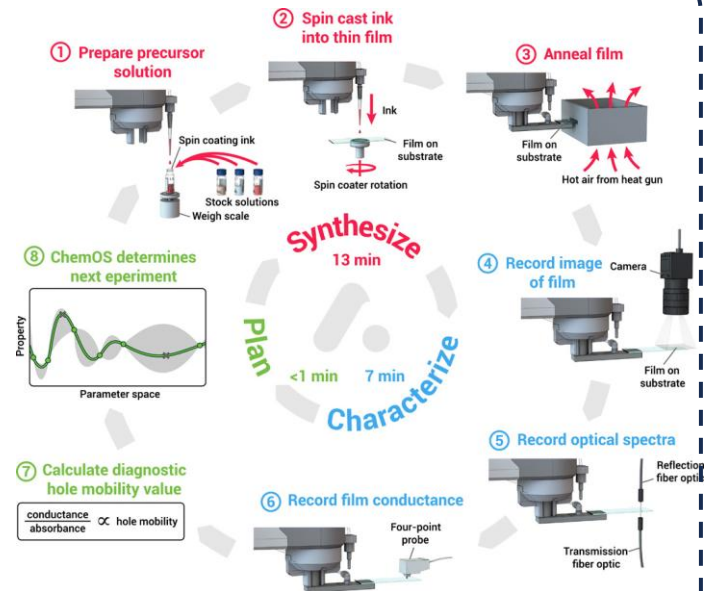
(1) 반도체 소자에 활용하기 위한 고유전율 박막 물질을 개발 용 Self-driving lab(SDL)과 workflow 설계

- (Requirement) High-throughput, 합리적인 footprint와 개발 비용/기간

Ada (UBC, thin-films)



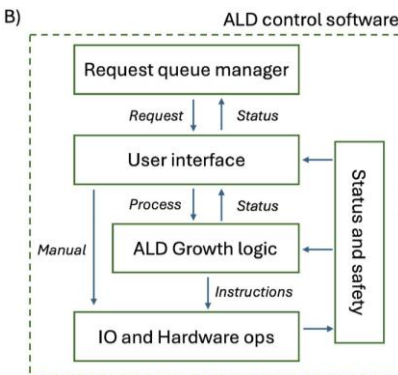
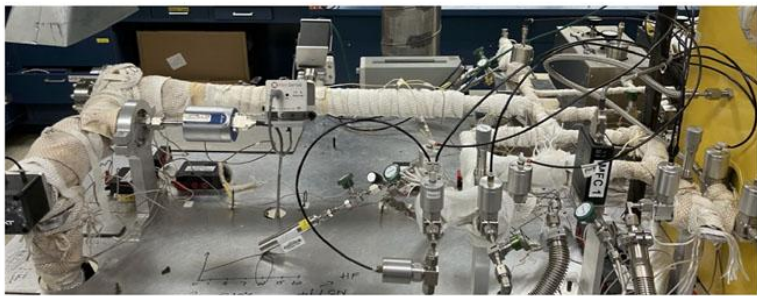
Nat Commun 13, 995 (2022)



Sci. Adv. 6, eaaz8867(2020)

Ada (Argonne Labs, ALD)

A)

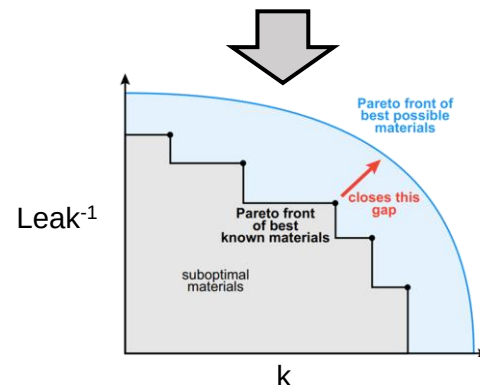
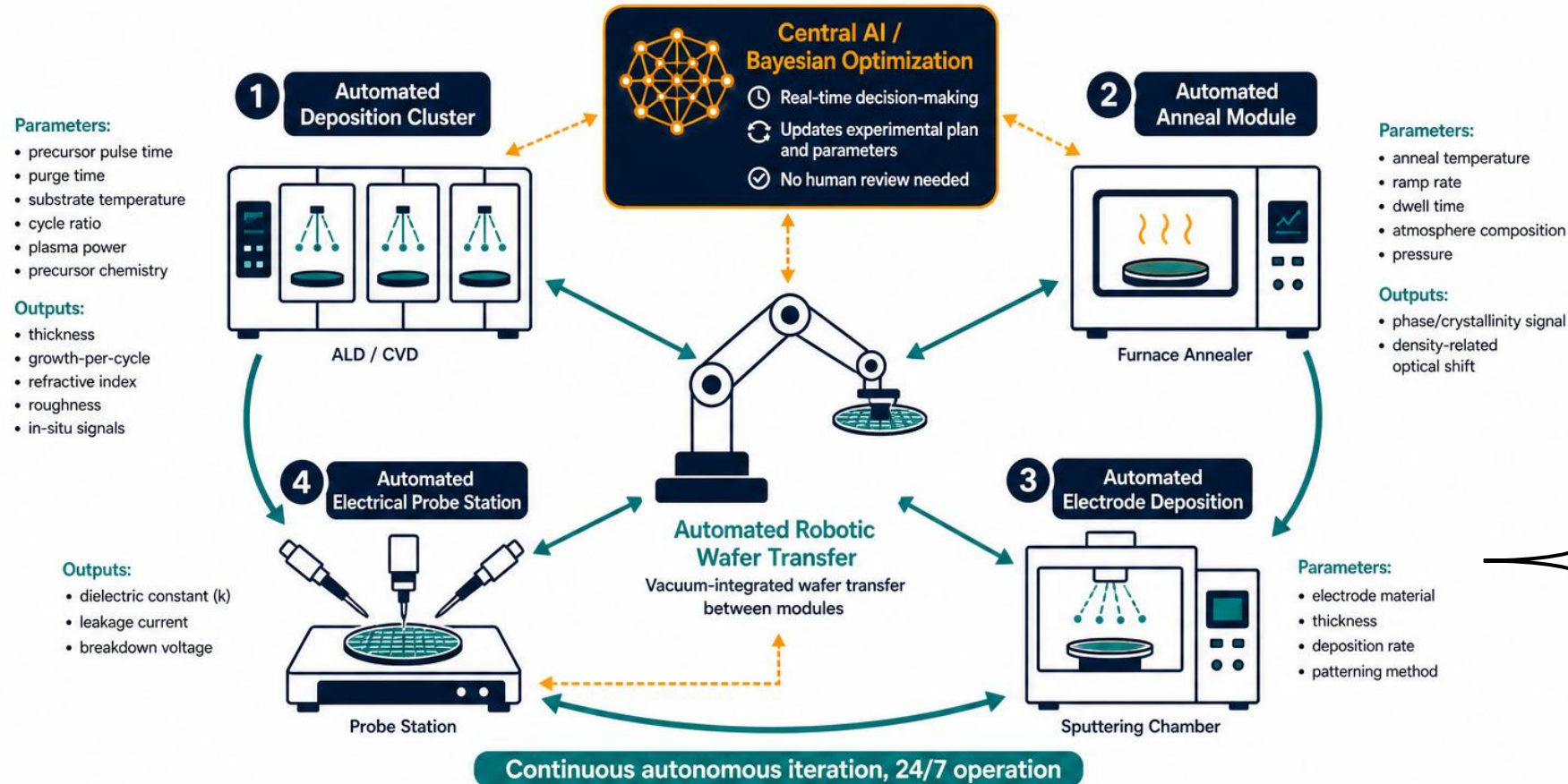


Rev. Sci. Instrum. 97, 053903 (2026)

Why it Falls Short

- Optimize **structural** or **optical** proxies (Raman signal, film conductivity, thickness, crystal phase)
— **not real device performance**
- Automate only **one process** stage (deposition/synthesis)
— **stop before anneal, electrode integration, or device-level testing**
- Assume structure-function **link** holds, without testing where it breaks down
- Gap gets critical for high-k dielectrics
— anneal and electrode steps transform the film after the "**fast**" measurement is taken
- High capital cost for narrow demonstrated scope (e.g., Ada: **~\$8M**, 3 years, one coating property)

Ideal SDL for high-k thin-films (Industrially capable)



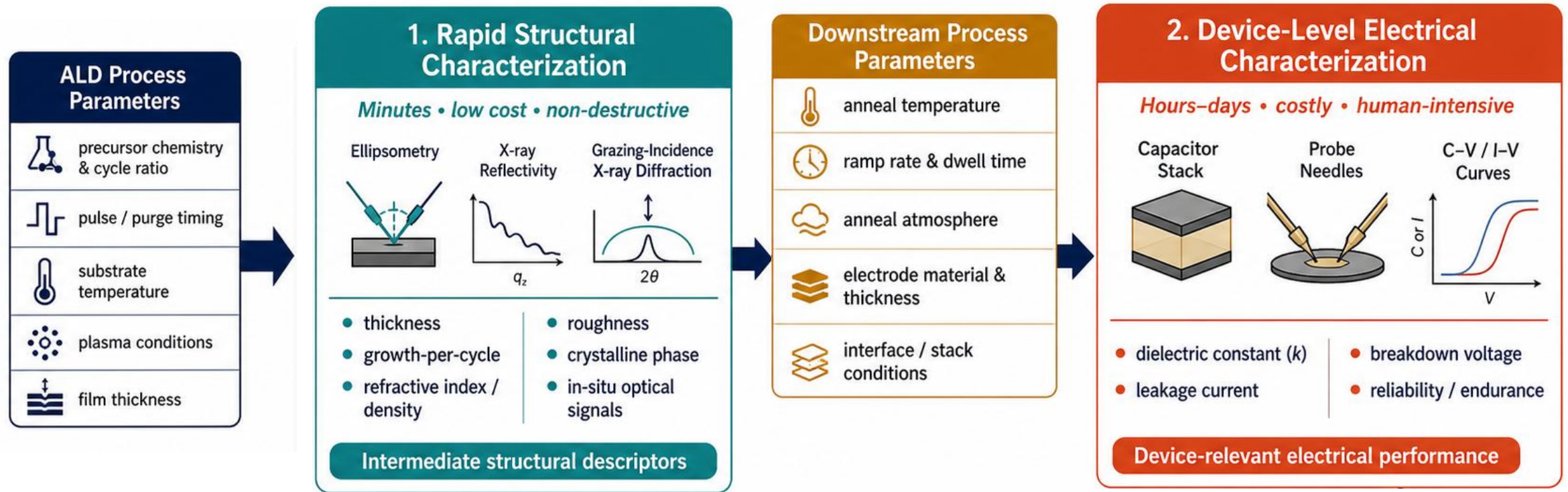
• **Expensive** — full automation of 4 stages (deposition, anneal, electrode, probe) needs custom robotics; Ada cost ~\$8M for just one step

• **Hard to automate** — anneal/electrode steps resist robotization; field itself flags multi-step handling as unsolved

• **Large footprint** — integrating 4 vacuum-compatible tools needs cleanroom space most labs don't have

• **Not necessary** — real bottleneck is the unproven fast-to-electrical correlation, not throughput; smart sampling solves this cheaper than full autonomy

Characterization **IS** the bottleneck



What We Need to Learn

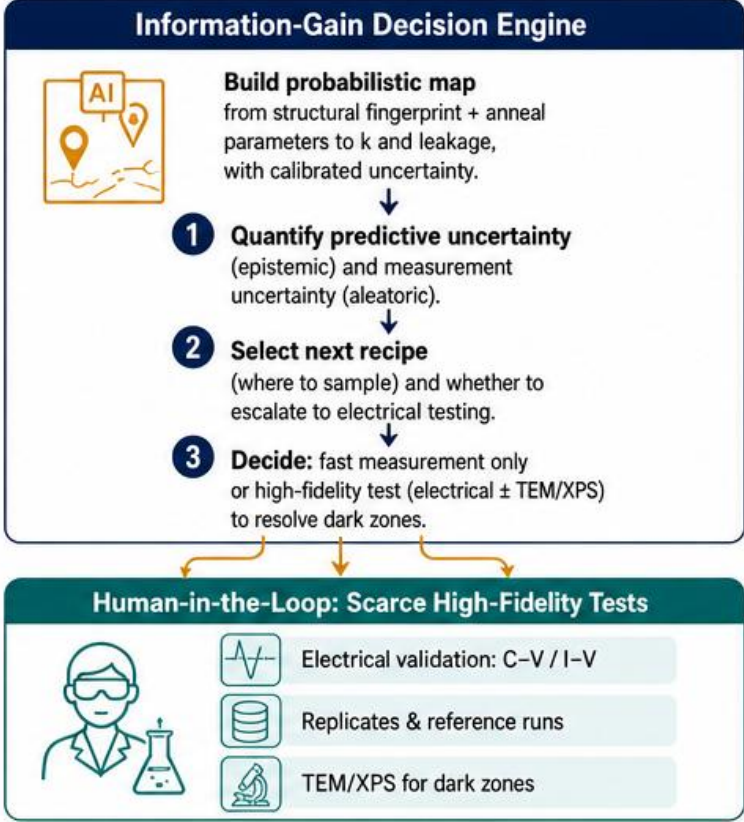
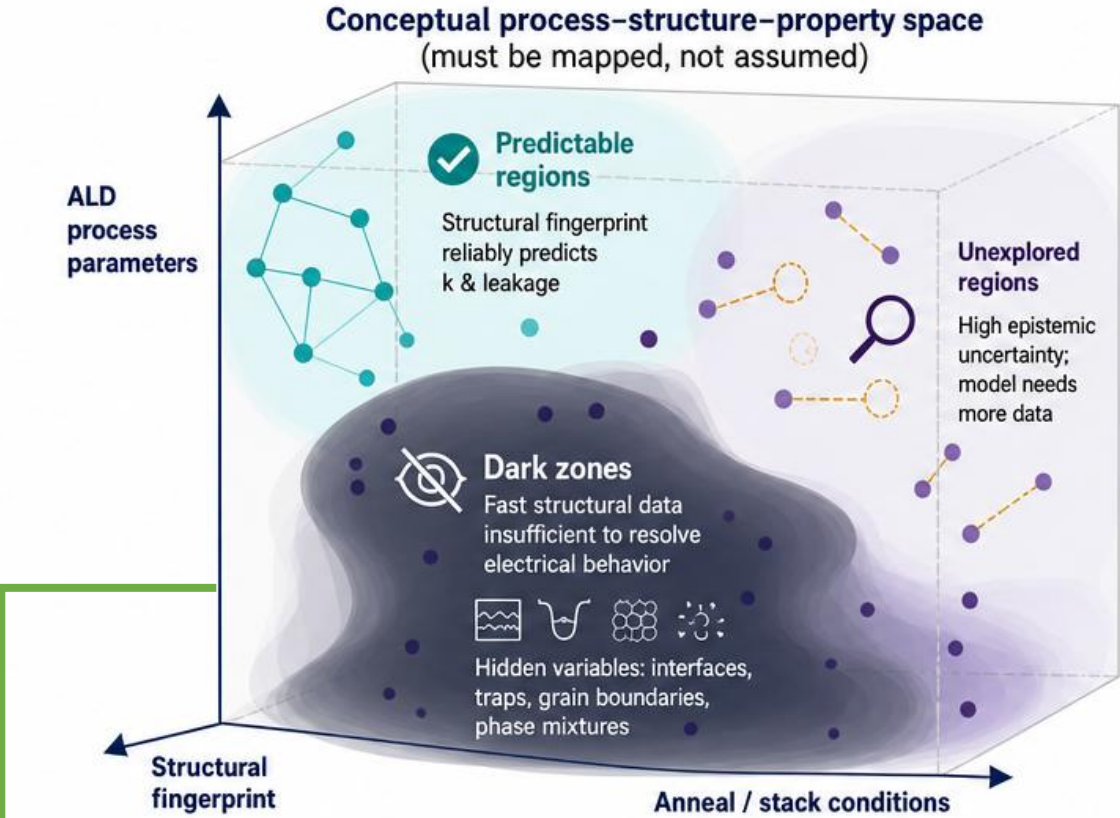
Structural fingerprint = $f(\text{ALD process parameters})$

Electrical performance = $f(\text{structural fingerprint}, \text{downstream process parameters})$

What if: we could predict electrical performance using structural fingerprints

ground-truth
feedback

Multi-Fidelity Active Learning That Knows What It Doesn't Know

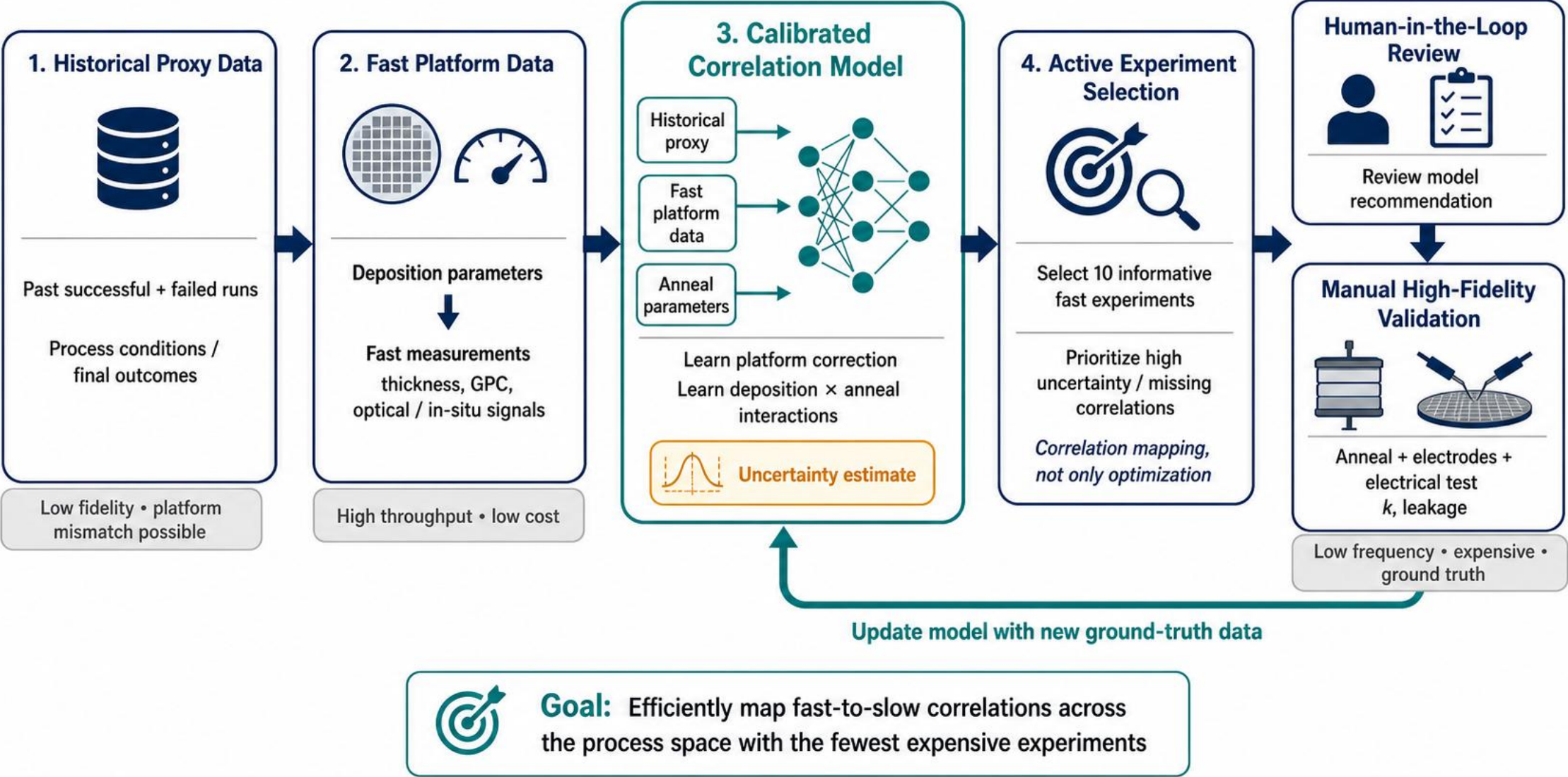


We are trying to find **regions** that we can **fastly** discover good performing materials

No need for multi-million automated characterization workflow

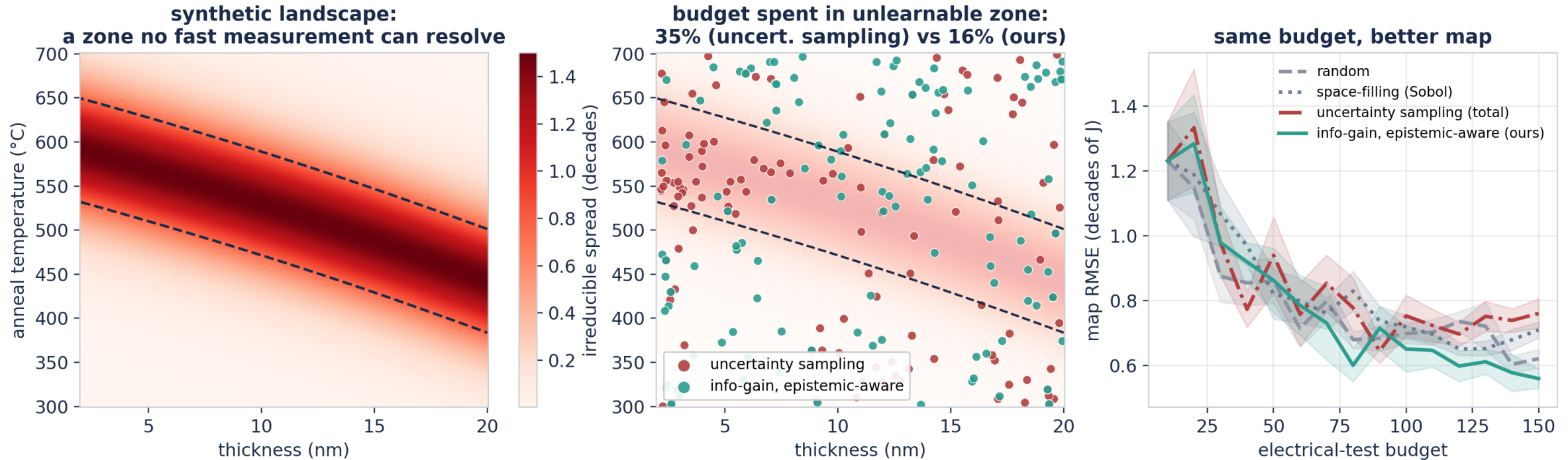
What we can do with this map?: We can **exploit** the **predictable regions** with **rapid (cheap) experiment** to discover **ACTUAL** materials that has good **electrical properties (without expensive experiments)**

Multi-Fidelity Learning Workflow for High-*k* Dielectrics



DEMONSTRATION

Proof of mechanism: the decision engine already works in silico



► **Result (6 campaigns, identical 150-test budget):** our policy reaches the baselines' final map accuracy with ~half the electrical tests, **wastes 16% vs 30–35%** of budget in the unlearnable zone, and detects that zone best

Key message Physics aside — the engine demonstrably learns what is learnable, flags what is not, and spends human time where it counts.

EXECUTION

A phased roadmap where every stage delivers standalone value

PHASE 0 · Months 0–6 Foundations

Data infrastructure & provenance schema
Retrospective modeling on existing ALD data
In-silico algorithm benchmark
Deliverable: validated decision engine + data backbone

PHASE 1 · Months 6–18 Human-in-the-loop SDL

Automated recipes + fast structural loop
Algorithm-selected electrical tests, human-executed
First structure→property map for one material system
Deliverable: virtual-metrology model v1 + mapped process window

PHASE 2 · Months 18+ Closing the loop

Automated sample handling & annealing integration
Expanded palette: dopants, laminates, electrodes
Transfer validation on device-representative stacks
Deliverable: autonomous platform + reusable materials dataset

Problem 2:

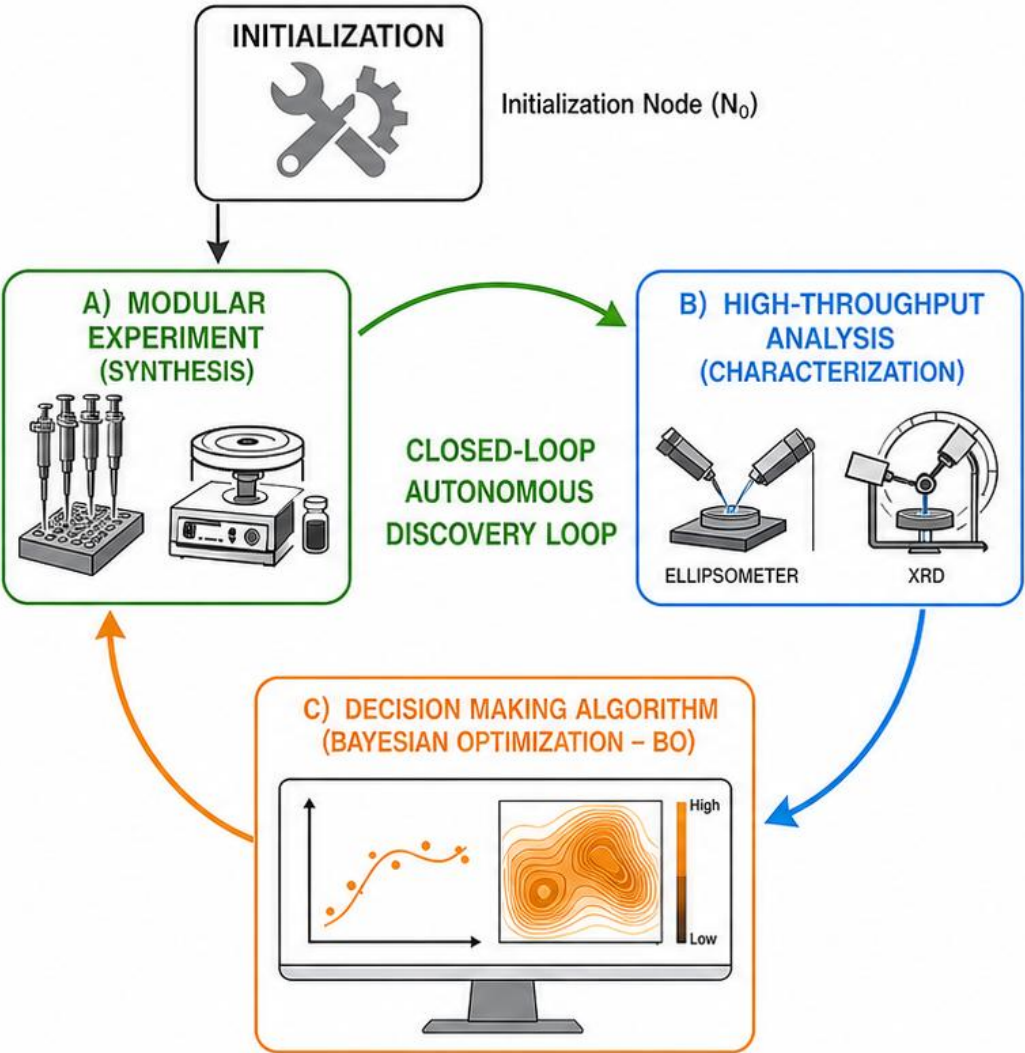
(2) 다양한 application의 물질 개발(박막, 무기, 유기; 반도체, 디스플레이, 배터리) 용 SDL을 가지고 있을 때, 이러한 SDL에 SW 개발/운영을 위한 범용 SW architecture 설계

•(Requirement)

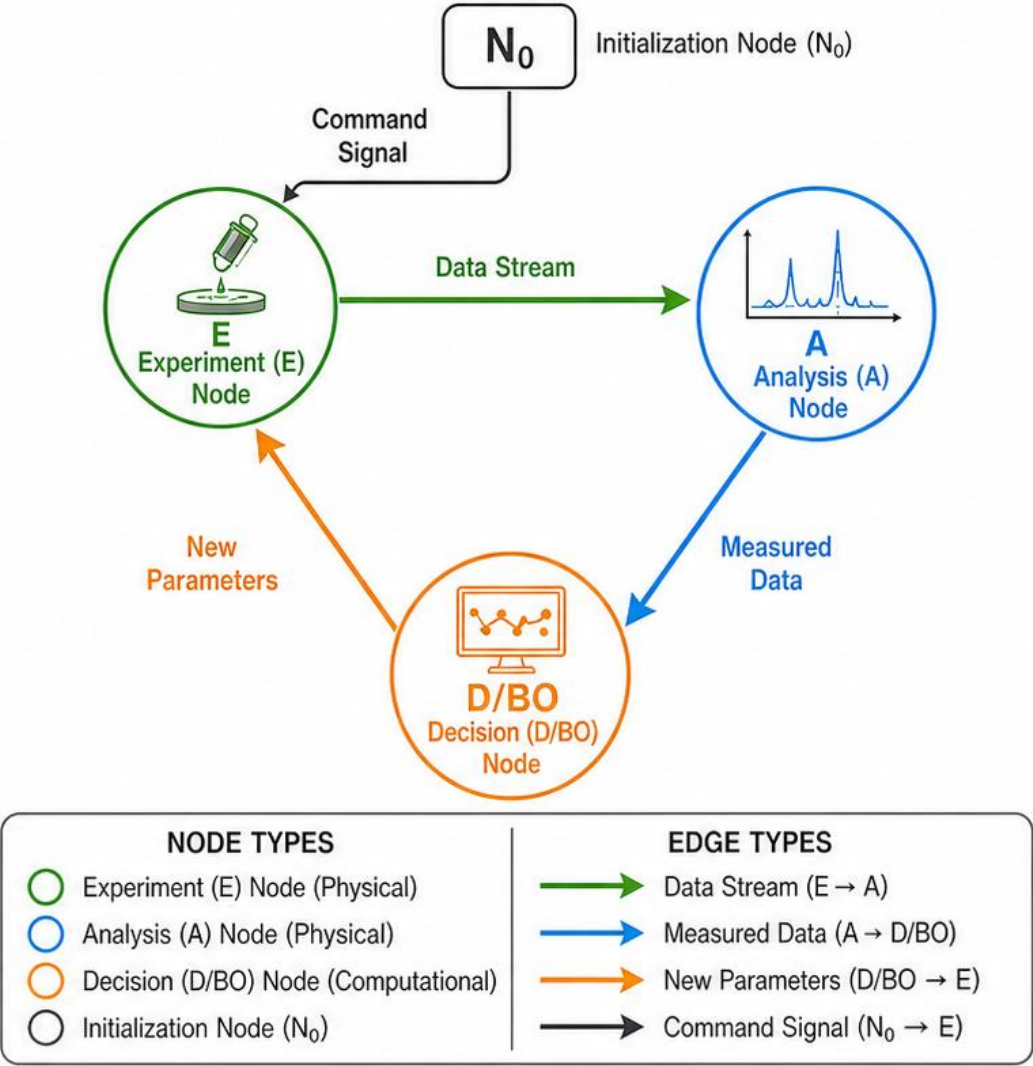
- 다양한 SDL 들을 개별적으로, 또한 통합적으로 운영할 경우에도 활용 가능
- SDL에 대한 전문 지식이 없는 물질/소재 연구자도 없이 쉽게 사용 가능
- 시스템을 구성하는 장치 변경(HW의 추가, 변경, 제거 등)에도 최소한의 burden으로 대응 가능

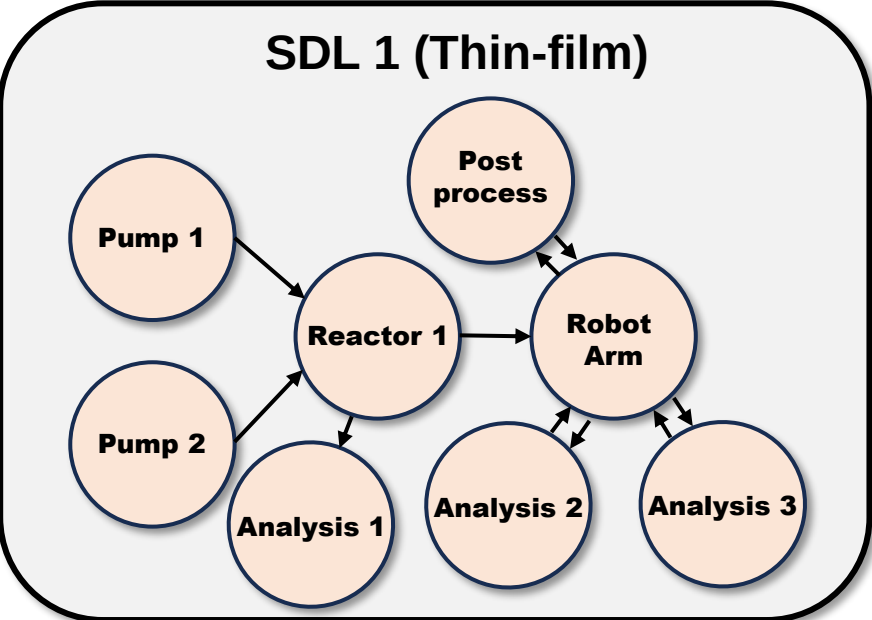
※ LLM이 아닌 본인의 idea의 이론/아이디어 제시

General Overview of an SDL



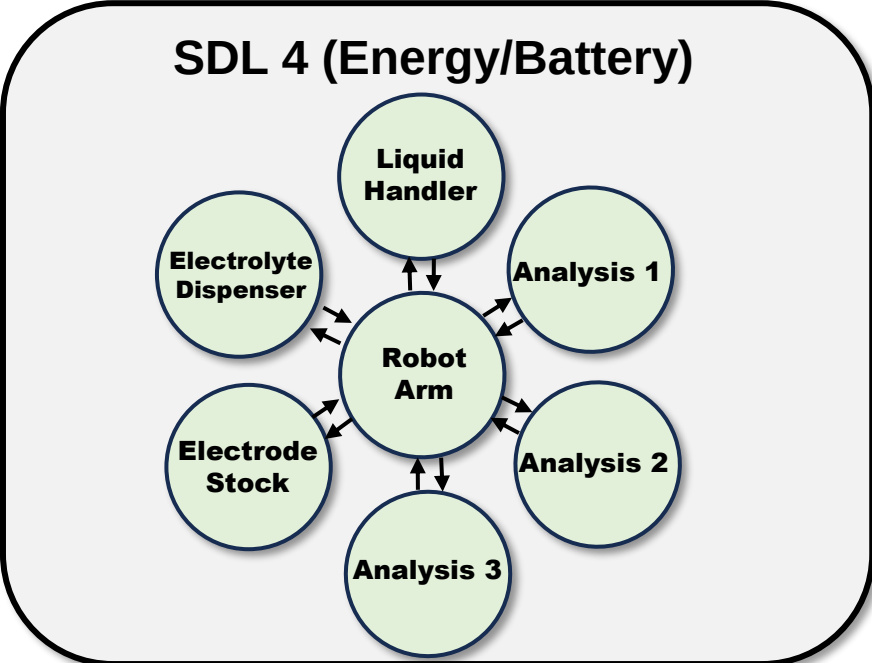
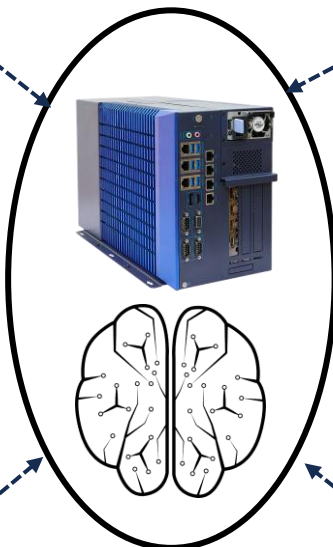
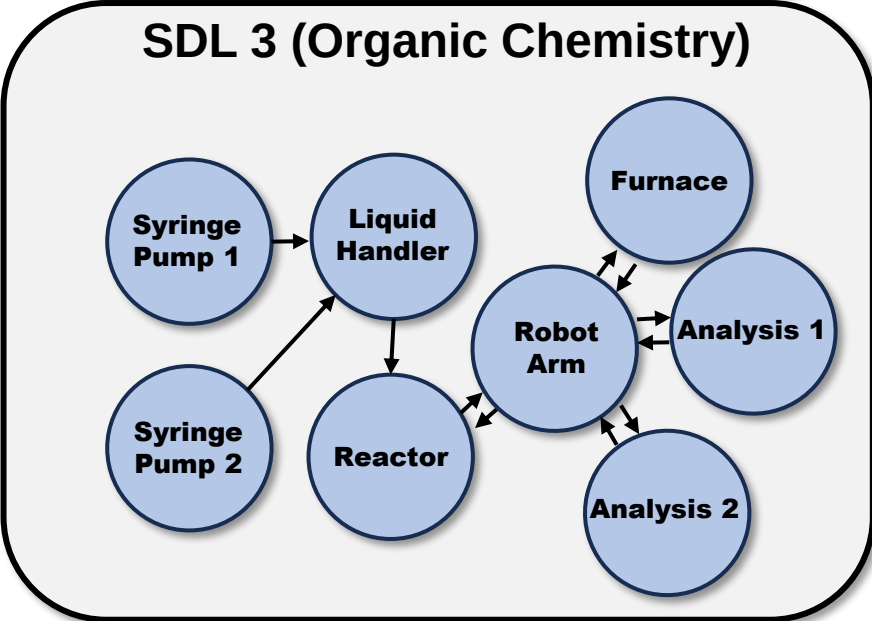
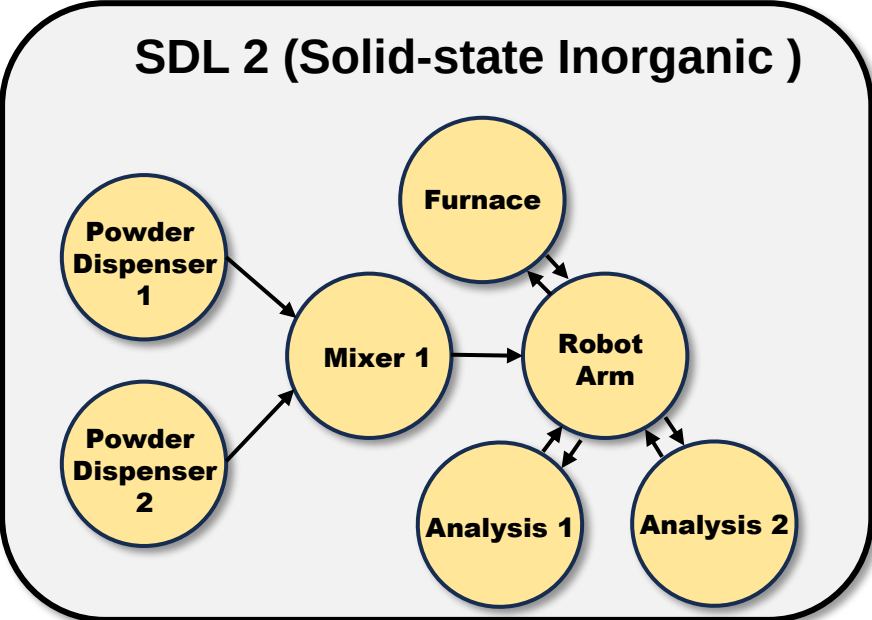
SDLs are somehow Graphs



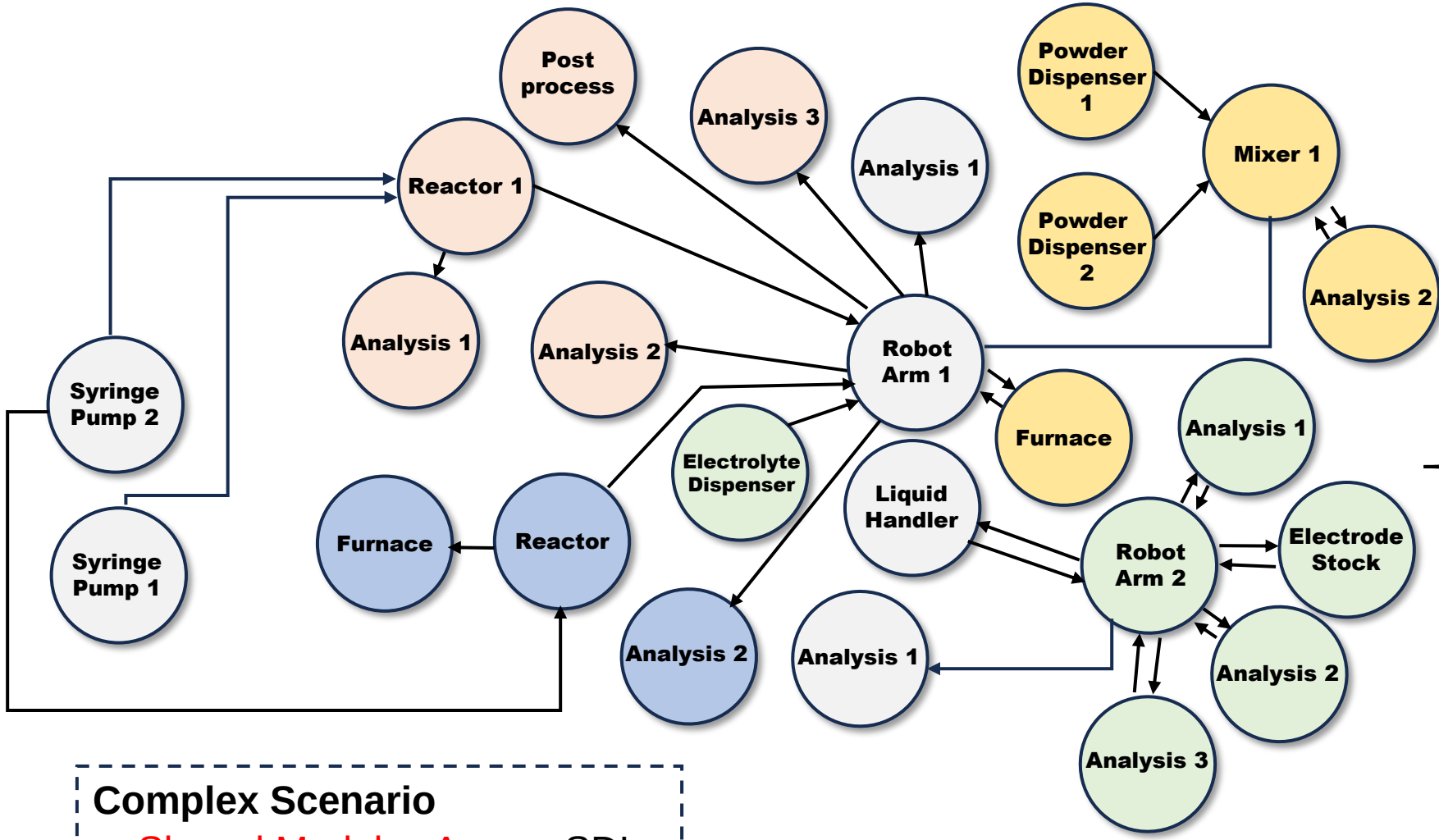


Simplest Scenario:

- Isolated SDLs
- No shared modules



Cluster of Interconnected SDLs with Shared Modules



Complex Scenario

- **Shared Modules Across** SDLs

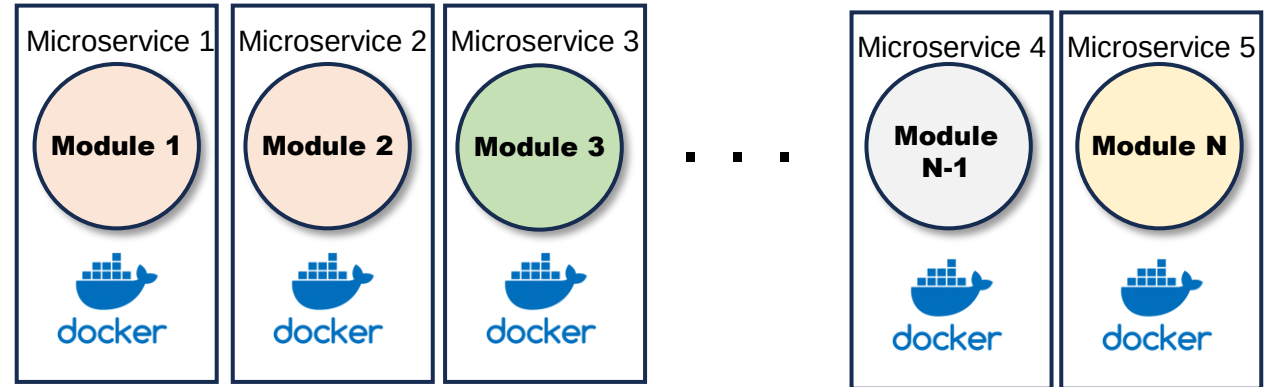
What we need

- Efficient **Control** of the SDLs and **real-time** monitoring
- Non-interruptive **scheduling** of the tasks
- Accessible to use by **diverse researchers** with limited control/coding skills
- Powerful workflow for **robust** and **simultaneous** usage of multiple SDLs at the same time
- Robust **data** management platform

My Idea

Modules:

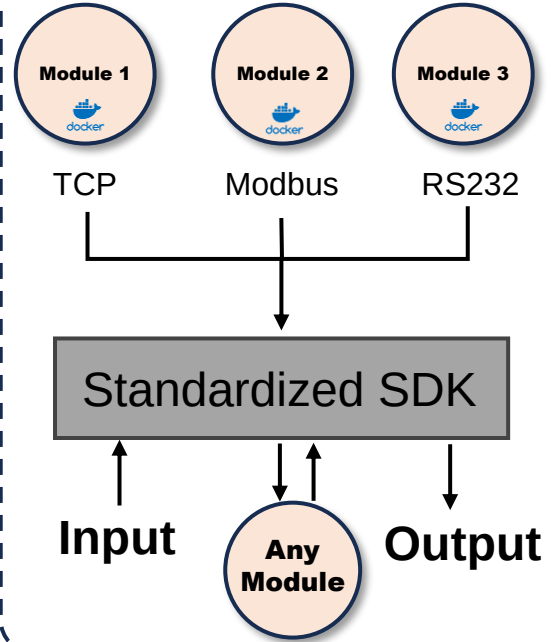
- 1- analysis instruments
- 2- Dispenser/pumps
- 3- Sensors
- 4- Robots
- 5- ...any component in SDL



Each  is an island 

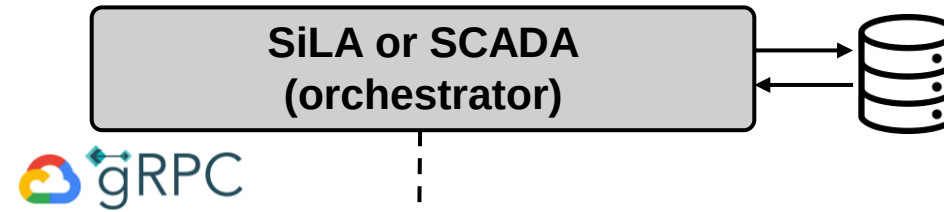
- 1. Works independently
- 2. Doesn't disturb or destroy other tasks
- 3. Safer workflows

Universal Communication Prot.



Steps:

- 1- Isolated modules → Microservices (MSs)
- 2- Define a universal grammar for MSs
- 3- a Flexible GUI for Multi-SDL control/monitoring
- 4- Bridge the hardware and software with orchestrator



React flow
Plotly.js
Typescript
WebSocket

GUI (Frontend)

- ① Drag-and-drop design of process
- ② Multi-SDL control/monitoring
- ③ Real-time Analysis
- ④ Easy Choice of DoEs (BO/LLM etc) and Task scheduling

Thank you